

MATH 708–2026. HOMEWORK 7. DUE ON APRIL 10.

- Unless otherwise stated, the exercises are from Voisin’s textbook.
- The homework problems may be submitted orally (during assigned office hours), in written/typed form, or as a combination of the two.
- You may submit any subset of problems totaling up to 100 points.
- If you submit more than 100 points’ worth, I will grade only the first 100 points you designate.
- Do not write anything on the cover sheet, as it will be used for grading. Bring it to office hours when you submit problems orally, and staple it to your written/typed solutions.

Problem 1. (5 points each part). Page 136, Exercise 1.

Problem 2. (10 points). Page 136, Exercise 2.

Problem 3. (5 points each part). Page 154, Exercise 1.

Problem 4. (5 points each part). Page 154, Exercise 2.

Problem 5. (10 points). Show that the map $W \subset V \mapsto \bigwedge^k W \subset \bigwedge^k V$ defines a holomorphic embedding of complex manifolds

$$\mathrm{Gr}(k, V) \hookrightarrow \mathbf{P}\left(\bigwedge^k V\right).$$

called the Plücker embedding. Deduce that $\mathrm{Gr}(k, V)$ is a projective variety.

Problem 6. (5 points). Let p be a primitive fifth root of unity. The group $G = \langle p \rangle \cong \mathbf{Z}/5\mathbf{Z}$ acts on $\mathbf{P}^3$ by $(z_0 : z_1 : z_2 : z_3) \mapsto (z_0 : pz_1 : p^2z_2 : p^3z_3)$. Describe all fixed points of this action on $\mathbf{P}^3$. Show that the surface $Y \subset \mathbf{P}^3$ defined by $z_0^5 + z_1^5 + z_2^5 + z_3^5 = 0$ is G -invariant and that the induced action of G on Y is fixed-point free. Conclude that the quotient $X = Y/G$ is a smooth compact complex surface. This surface is called a *Godeaux surface*.

Problem 7. (5 points). Let L be a holomorphic line bundle on a complex manifold X . Show that $R(X, L) := \bigoplus_{m \geq 0} H^0(X, L^{\otimes m})$ has a natural ring structure. For example, the canonical ring of X is defined by

$$R(X, \omega_X) := \bigoplus_{m \geq 0} H^0(X, \omega_X^{\otimes m}).$$

It is a famous theorem of Birkar, Cascini, Hacon, and McKernan that this ring is finitely generated for every projective manifold X .

Problem 8. (5 points). Let D be a smooth hypersurface in a complex manifold X . Show that

$$\omega_D \simeq \omega_X(D)|_D.$$

Hint: Use the definition of the normal bundle.

Problem 9. (5 points each part). Let L be a holomorphic line bundle on a compact complex manifold X .

- (a) Show that the vector space $H^0(X, L)$ of global holomorphic sections of L is finite-dimensional.
- (b) A point $x \in X$ is called a *base point* of L if $s(x) = 0$ for all $s \in H^0(X, L)$. The *base locus* $\text{Bs}(L)$ is the set of all base points of L . Choose a basis $s_0, \dots, s_N \in H^0(X, L)$. Show that

$$\text{Bs}(L) = Z(s_0) \cap \dots \cap Z(s_N),$$

and deduce that $\text{Bs}(L)$ is an analytic subvariety of X .

- (c) Show that we have a holomorphic map $\varphi_L : X \setminus \text{Bs}(L) \rightarrow \mathbf{P}^N$ given by $x \mapsto [s_0(x) : \dots : s_N(x)]$.
- (d) Show that $\varphi_L^* \mathcal{O}_{\mathbf{P}^N}(1) \simeq L|_{X \setminus \text{Bs}(L)}$.

Problem 10. (5 points each part). Let $\mathbf{P}^n$ be complex projective space, and let $L = \mathcal{O}_{\mathbf{P}^n}(k)$ be the holomorphic line bundle of degree k .

- (a) Show that any nonzero homogeneous polynomial

$$s \in \mathbf{C}[z_0, \dots, z_n]_k$$

of degree k defines a nonzero global holomorphic section of L .

- (b) Show that the resulting map $\mathbf{C}[z_0, \dots, z_n]_k \rightarrow H^0(\mathbf{P}^n, L)$ is linear and injective.
- (c) Show that this map is an isomorphism. *Hint:* Let $t \in H^0(\mathbf{P}^n, L)$ be a nonzero section, choose a nonzero section $s_0 \in H^0(\mathbf{P}^n, L)$ coming from a homogeneous polynomial of degree k , and consider the meromorphic function $F = t/s_0$. Pull it back to $\mathbf{C}^{n+1} \setminus \{0\}$, multiply by s_0 , and use Hartogs' theorem to extend across the origin.
- (d) Let $L = \mathcal{O}_{\mathbf{P}^n}(1)$. Using a basis of $H^0(\mathbf{P}^n, L)$, show that the associated map $\varphi_L : \mathbf{P}^n \rightarrow \mathbf{P}(H^0(\mathbf{P}^n, L)^*)$ is a holomorphic embedding.

Problem 11. (10 points). Let X be a Kähler manifold. Show the following identities:

$$[\bar{\partial}, L] = [\partial, L] = [\bar{\partial}^*, \Lambda] = [\partial^*, \Lambda] = 0.$$

Also prove that $[\bar{\partial}^*, L] = i\partial$, $[\partial^*, L] = -i\bar{\partial}$.

Problem 12. (10 points). Let X be a Kähler manifold. Show that the operators L, d, d^* acting on $A^*(X)$ determine the complex structure of X .

Problem 13. (10 points). Let X be a complex manifold endowed with a Kähler form ω , and let α be a closed $(1, 1)$ -form such that $\Lambda\alpha = 0$. Show that α is harmonic, i.e. $\Delta\alpha = 0$.

Problem 14. (10 points). Let X be a Kähler manifold. Show that the Kähler form ω is harmonic.

Problem 15. (10 points). Let X be a compact Kähler manifold. Show that every global holomorphic p -form is harmonic with respect to any Kähler metric on X .

Problem 16. (5 points). Show that $H^{p,q}(\mathbf{P}^n) = 0$ unless $p = q$. Show moreover that for $0 \leq p = q \leq n$, the space $H^{p,p}(\mathbf{P}^n)$ is one-dimensional.